

SATQUERY AI

COMPREHENSIVE PLATFORM WALKTHROUGH & ARCHITECTURE REPORT

Grounded Earth Observation Intelligence · Multi-Modal Sensor Fusion · Anti-Hallucination Core
Version: 0.4.0-Production | Authority: ISRO/SAC & Remote Sensing Decision Suite | Date: August 2026

EXECUTIVE MISSION STATEMENT: *SatQuery AI is a space-data intelligence platform designed to eliminate model hallucination in Earth Observation. Operating strictly under the core directive: 'Never guess when the available data cannot support the answer', the system dynamically assesses input modalities, validates external telemetry requirements, delineates physical ground-truth disaster sectors, and generates official decision-ready briefings for government departments.*

1. Multi-Modal Satellite Ingestion & Modality Intelligence

The platform supports single-scene and dual-scene comparative acquisitions across major spaceborne sensor formats:

- **Supported Formats:** GeoTIFF (.tif/.tiff), PNG, JPEG with preservation of multi-spectral bands.
- **Scene A & Scene B Dual Slots:** Clearly demarcates baseline reference imagery (Scene A) from post-event acquisitions (Scene B).
- **Automatic Modality Intelligence:** Eliminates manual sensor selection by analyzing pixel variance and spectral saturation to automatically identify *Optical / Multi-Spectral* vs. *SAR Radar Imagery* (Sentinel-1 C-Band).
- **Image Relationship Engine:** Automatically detects whether dual uploads represent a *Bi-Temporal Pair* (same sensor, different time), a *Cross-Modal Pair* (Optical + SAR), or flags a *Geographic Mismatch*.

2. Anti-Hallucination & Quad-Tier Evidence Categorization

Traditional Vision-Language Models often hallucinate speculative answers when asked unanswerable questions (e.g. predicting future flood arrival from a single static photograph). SatQuery AI strictly categorizes all output into four rigorous tiers:

Category	Definition	Platform Behavioral Rule
1. Observed	Direct physical evidence verifiable in image pixels (e.g. water bodies, building rooftops).	Stated with high confidence as physical fact.
2. Inferred	Contextual assessments derived from physical proximity, terrain gradients, and heuristics.	Explicitly labeled as AI inference / probabilistic estimate.
3. Predicted	Future projections requiring multi-source temporal feeds (rainfall, river gauges).	BLOCKED if required telemetry feeds are missing.
4. Missing Data	Explanations detailing precisely why the query cannot be answered with current inputs.	Recommends exact sensor / telemetry feeds needed.

3. 'Do I Have Enough Evidence?' 5-Point Data Verification

Before executing any analytical workflow, SatQuery AI checks a 5-point data requirement matrix:

1. **Satellite Imagery Available:** Validates raster dimensions, bands, and resolution.
2. **Temporal Pair Status:** Verifies if comparative questions have both Before and After scenes.
3. **Multi-Source Auxiliary Telemetry:** Checks rainfall gauges, DEM elevation, and meteorological forecasts.

- 4. **Spatial Coverage Compatibility:** Verifies geographic overlap between scenes.
- 5. **False-Positive Exclusion Calibration:** Ensures baseline permanent water bodies (rivers/reservoirs) are not misclassified as active disaster inundation.

4. Disaster-Area Detection & Interactive Visual Grounding

The visual grounding engine transforms raw classification into physical, inspectable geographic overlays:

- **Interactive Overlays:** Renders semi-transparent vector polygon boundaries and bounding boxes directly on top of satellite imagery without altering raw pixels.
- **Region Delineation:** Identifies distinct affected sectors (Region 01, Region 02, Region 03) with calculated square kilometer footprints and percentage of region of interest (ROI).
- **'Why Was This Area Marked?' Inspector:** Clickable pins provide deep evidence rationale, explaining specular radar backscatter attenuation (< -18 dB), modified NDWI spectral drops, or delta-NBR burn scar signatures.
- **Multi-Disaster Support:** Specialist models for Flood & Inundation, Forest Fire / Wildfire Burn Scars, Landslide Slope Failure, and Cyclone Debris.
- **Bi-Temporal Delta & Swipe:** Interactive Before (T0) vs After (T1) comparative view for instantaneous visual change verification.

5. Cross-Modal Consensus & Multi-Temporal Change Stories

- **Cross-Modal Agreement (Optical + SAR):** When both optical and radar scenes are provided, SatQuery computes an agreement consensus score (e.g. *92% High Agreement*). Disagreements caused by optical cloud cover or radar terrain shadows are explicitly flagged rather than concealed.
- **Chronological Change Story:** Replaces simple 'Before/After' outputs with an evolutionary narrative timeline tracking land cover transitions across 5 domains (Urban Growth, Deforestation, Water-Body Shifts, Infrastructure, Agriculture).

6. Government Decision Mode (Multi-Agency Operational Suite)

Dedicated operational frameworks tailored for Indian space and disaster management authorities:

Agency / Sector	Authority	Primary Operational Inquiries
■ Disaster Management	NDEM / NRSC-ISRO	Which areas are inundated? Where has flood spread? Submerged transit corridors.
■ Forest Monitoring	Forest Survey of India	Has canopy density decreased? Where are active burn scars / deforestation zones?
■■ Urban Planning	MoHUA / Smart Cities	Has built-up area expanded? Where has new construction encroached on green belts?
■ Agriculture	Mahalanobis Crop Centre	Has crop vigor changed? Which parcels exhibit drought or waterlogging stress?
■ Water Resources	Central Water Commission	Has surface water coverage expanded/contracted? Reservoir storage volume estimates.

7. Production-Grade Authentication & Workspace Isolation

A secure, enterprise-grade access control and user workspace isolation engine:

- **Security Architecture:** Passwords hashed with **Bcrypt (12 rounds)**; zero plaintext passwords. Stateless **JWT sessions (HS256)** with Bearer token authentication.

- **Strict User Isolation:** Relational SQLite/PostgreSQL schema tying analyses and reports to user_id. User A can never access or query User B's historical records (enforced via 404 guards).
- **Workspace Features:** Interactive registration with live 5-point password strength meter, email verification, password reset tokens, personal dashboard metrics, searchable history repository, and profile management with danger zone account deletion.
- **Hybrid Offline Resilience:** Automatic client-side fallback storage engine ensuring seamless registration, login, analysis, and report generation even on standalone Vercel preview deployments.

8. Official Decision Briefing Generator & Vercel Deployment

- **Decision-Ready Briefing Export:** Generates official government briefings titled SatQuery AI — Satellite Disaster Assessment formatted with reference IDs, classification banners (FOR OFFICIAL USE ONLY), evidence audits, and print-ready PDF styling.
- **Cloud & Vercel Deployment:** Fully configured with vercel.json SPA routing rewrites (preventing 404s on refreshes) and pushed to GitHub (<https://github.com/juwairiyah2010/satquery-ai>).

SatQuery AI Platform Architecture Report · Verified & Generated via Autonomous Space-Data Engineering Core · Confirmed Zero-Hallucination Grounding